

| Qn      | Suggested Solutions                                                                                                                            |
|---------|------------------------------------------------------------------------------------------------------------------------------------------------|
| 1(a)(i) | $\Delta p = mv - mu$<br>$= 0.140(-4.0 - (5.4))$                                                                                                |
|         | $\Delta p = -1.3 \text{ N s}$                                                                                                                  |
| (ii)    | $F = \left  \frac{\Delta p}{\Delta t} \right  = \frac{1.3}{0.04} = 32.5 \text{ N}$                                                             |
|         | $F = N - mg$<br>$32.5 = N - 0.14(9.81)$<br>$N = 34 \text{ N}$                                                                                  |
|         | By Newton's 3 <sup>rd</sup> law,<br>magnitude of the force exerted by the ball on the bar = magnitude of the force by the bar on the ball, $N$ |
| (b)     | Taking moments about B,                                                                                                                        |
|         | $34(75) + 0.450(9.81)(25) = F_A(20)$                                                                                                           |
|         | $F_A = 133 \text{ N}$                                                                                                                          |

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H2 (9749) Physics

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